

AI-SA-IN-TH-000018

Module 8.1 · LLM Evaluation Fundamentals · AI Safety: Evaluation, Guardrails & Responsible Deployment

Grounding and Factual Verification

RAG, Citation Enforcement, Chain-of-Verification, RAGAS, and Production Grounding Architecture

9 min

Duration

B3 – Apply

Bloom's

L3 – Advanced

Level

Theory

Domain

Learning Objectives

- 1 Define grounding and explain why it reduces hallucination without eliminating it
- 2 Describe the five-step RAG grounding pipeline and identify which steps affect faithfulness
- 3 Design a citation-enforced prompt template that forces the model to attribute claims to sources
- 4 Apply the Chain-of-Verification (CoVe) algorithm to improve factual accuracy on a given query
- 5 Compute a RAGAS faithfulness score from a set of claim-level NLI judgements
- 6 Compare grounding strategies across five dimensions and select the appropriate approach for a given domain

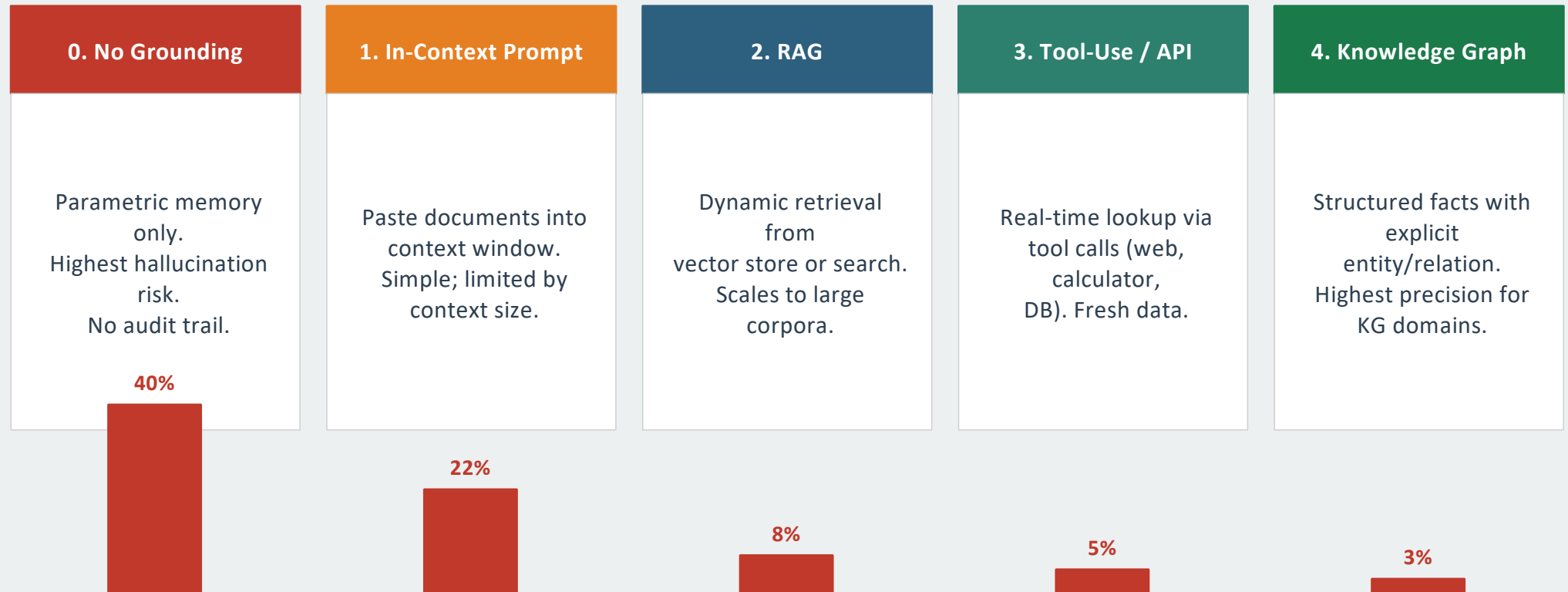
What Is Grounding?

Grounding

Grounding is the practice of constraining an LLM's outputs to information that is explicitly present in, or directly supported by, a verified external source — preventing the model from relying solely on memorised parametric knowledge when generating factual claims.

- Without grounding: model uses parametric knowledge (memorised during training) → high hallucination risk, knowledge cutoff limitations, no audit trail
- With grounding: model generates claims it can attribute to a provided document, database, or tool result → lower hallucination rate, auditable citations
- Grounding reduces, does not eliminate, hallucinations: models can still misrepresent grounding documents (faithfulness hallucination) even with perfect retrieval
- Grounding is not the same as retrieval: retrieval finds relevant documents; grounding constrains generation to those documents — both are necessary
- Measurable: grounding quality is measured by RAGAS faithfulness score and NLI entailment rate — target faithfulness > 0.90 for high-stakes deployments

Grounding Architecture Spectrum



Approx. hallucination rate ↑

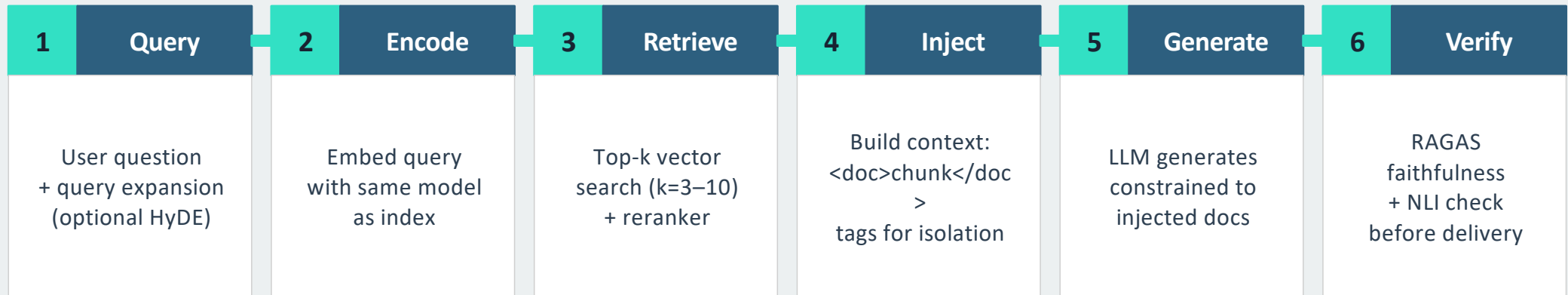
RAG as the Primary Grounding Mechanism

Retrieval-Augmented Generation (RAG)

RAG reduces hallucination by retrieving relevant documents from a verified corpus at query time and injecting them into the model's context window, constraining the model to generate claims that are attributable to the retrieved evidence.

- Hallucination reduction: RAG reduces domain hallucination rates by 5–7× by replacing reliance on potentially incorrect parametric memory with verified external text
- Two failure modes remain: (1) retrieval failure — wrong or missing documents retrieved; (2) faithfulness failure — model generates claims not supported by the retrieved documents
- Faithfulness vs accuracy: high retrieval accuracy is necessary but not sufficient — faithfulness scoring measures whether the model stayed within the retrieved context
- Chunk size matters: too-small chunks lose context; too-large chunks dilute relevant signal — optimal chunk size is domain-dependent (150–400 tokens typical)
- Knowledge cutoff bypass: RAG allows the model to answer about post-training-cutoff events by retrieving current documents — essential for time-sensitive deployments

RAG Grounding Pipeline — Retrieval → Injection → Generation



Step 4 (Inject) is the critical grounding step — XML tagging isolates document content from instructions, preventing indirect injection.

Step 6 (Verify) closes the loop — faithfulness scoring catches step-5 failures before the response reaches the user.

Citation-Enforced Generation

Citation Enforcement

A prompt engineering strategy that requires the model to attribute every factual claim to a specific source document using inline markers — making faithfulness auditable and providing users with evidence to verify claims independently.

- Mechanism: prompt includes numbered retrieved documents [1], [2], [3]... and instructs the model to cite [n] for every factual claim
- Key benefit: unverifiable claims become visible — if the model cannot cite a source, it should not make the claim
- Faithfulness improvement: citation enforcement reduces extrinsic hallucinations by 60–80% by forcing explicit source attribution
- User benefit: citations enable users to verify claims and read primary sources — especially important in medical, legal, and research contexts
- Limitation: models can hallucinate citation numbers (cite [2] for a claim actually from [1]) — verify citation accuracy with post-processing NLI check

Citation Enforcement Prompt Template

Python / Prompt

```
CITATION_SYSTEM = """You are a research assistant. Answer the question using ONLY the documents provided below.
```

```
RULES:
```

1. Every factual claim MUST be followed by [N] citing the document number.
 2. If a claim is NOT supported by any document, write: [UNSUPPORTED]
 3. Do NOT use your own knowledge – only information in the documents.
 4. If the documents do not contain the answer, say so explicitly.
- ```
"""
```

```
def build_grounded_prompt(question: str, docs: list[dict]) -> str:
 doc_text = '\n'.join(
 f'[{i+1}] {d["title"]}\n{d["content"]}' for i,d in enumerate(docs)
)
 return f'{CITATION_SYSTEM}\n\nDocuments:\n{doc_text}\n\nQuestion: {question}'
```

Parse citations with regex `r'\[(\d+)\]'` to verify each citation number is within range `[1..len(docs)]`. Out-of-range citations signal fabricated references.



# Chain-of-Verification (CoVe) Prompting

## Chain-of-Verification (Dhuliawala et al., Meta, 2023)

CoVe is a four-step prompting framework that reduces LLM hallucinations by having the model generate an initial response, identify its own factual claims, independently verify each claim, and then produce a revised response correcting any identified errors.

- Step 1 (Draft): generate initial response to the question
- Step 2 (Plan verifications): instruct the model to list all factual claims in the draft as explicit verification questions ('What year was X founded?', 'Who invented Y?')
- Step 3 (Execute verifications): answer each verification question independently — without the draft in context — to avoid anchoring bias
- Step 4 (Final response): produce revised response that corrects any claims that failed verification
- Hallucination reduction: CoVe reduces factual errors by 28–40% compared to direct prompting on list-based factual tasks (Wikidata, FactScore benchmark)
- Cost: 3–4 additional LLM calls per query — justified for high-stakes domains; too expensive for high-volume consumer applications

# CoVe — Four-Step Worked Example

Query: 'Tell me about Marie Curie's Nobel Prizes'

## Step 1: Draft Response

Draft: 'Marie Curie won two Nobel Prizes — the Nobel Prize in Physics in 1903 (shared with Pierre Curie and Henri Becquerel) and the Nobel Prize in Chemistry in 1911. She was the first woman to win a Nobel Prize and the only person to win Nobel Prizes in two different sciences.'

## Steps 2–4: Verification and Correction

Step 2 — Verification questions generated:

- Q1: What year did Curie win Nobel Physics?
- Q2: Who shared the Physics prize with her?
- Q3: What year did Curie win Nobel Chemistry?
- Q4: Was she the first woman to win a Nobel?

Step 3 — Independent answers (no draft in context):

- Q1: 1903 ✓ Q2: Pierre Curie and Henri Becquerel ✓
- Q3: 1911 ✓ Q4: Yes, first woman to win Nobel ✓

Step 4: No corrections needed — draft verified. Final response = draft.

If Step 3 had returned 'Becquerel' only (omitting Pierre), Step 4 would have corrected the attribution.

# Grounding Methods — Comparison

| Method               | Hallucination Reduction | Latency         | Cost         | Knowledge Freshness          | Best For                              |
|----------------------|-------------------------|-----------------|--------------|------------------------------|---------------------------------------|
| In-context documents | ~60%                    | 0ms overhead    | 0            | Static (context-provided)    | Short docs; one-off queries           |
| RAG (vector search)  | ~75–80%                 | 50–200ms        | Low          | Updated with index           | Document QA; enterprise knowledge     |
| Citation enforcement | ~70–80%                 | 0ms overhead    | 0            | As current as retrieved docs | All RAG; verifiable outputs required  |
| CoVe prompting       | ~28–40%                 | 3–4× query cost | High         | Training cutoff              | High-stakes; low-volume factual tasks |
| Tool use (web/API)   | ~85–90%                 | 200–500ms       | Medium       | Real-time                    | Current events; dynamic data; math    |
| Knowledge graph      | ~88–93%                 | 100–300ms       | High (setup) | As current as KG update      | Structured entity-relation queries    |

# RAGAS Evaluation for Grounded Systems

## RAGAS Production Evaluation

RAGAS is run as the standard quality gate for grounded RAG deployments — before each production release, a representative evaluation set of 100–500 question-context-answer triples is scored across all four metrics. Any metric below threshold blocks deployment.

- Evaluation set construction: collect real user queries from production logs or create representative synthetic questions; ensure coverage of all topic categories
- Minimum viable score before production: Faithfulness  $\geq 0.90$ , Answer Relevancy  $\geq 0.85$ , Context Precision  $\geq 0.75$ , Context Recall  $\geq 0.80$
- Regression detection: compare each new model version's scores against the baseline — any metric dropping  $> 0.05$  blocks deployment
- Continuous monitoring: run RAGAS on a 5% sample of production queries daily — alert if faithfulness falls below 0.85 (potential retrieval or model regression)
- RAGAS + NLI together: RAGAS flags potential hallucinations at the query level; NLI provides sentence-level attribution — use both in production for complete coverage

# RAGAS Faithfulness Score — Worked Calculation

$$\text{Faithfulness} = |\text{Supported Claims}| / |\text{Total Claims in Answer}|$$

1. Answer: 'Metformin is the first-line T2D treatment [✓], recommended by ADA [✓], FDA-approved in 1994 [?], dosed at 500mg twice daily [?], reduces HbA1c by 1.5% [?]'
2. Step 1 — Extract atomic claims: 5 claims extracted (each sentence/clause is one claim)
3. Step 2 — NLI check each claim against retrieved context (BNF excerpts):
4. Claim 1 (first-line): ENTAILED ✓ Claim 2 (ADA): ENTAILED ✓
5. Claim 3 (1994): NEUTRAL ✗ Claim 4 (500mg): NEUTRAL ✗ Claim 5 (1.5%): NEUTRAL ✗
6. Step 3 — Faithfulness = 2 entailed / 5 total = 0.40 → BLOCK — hallucination detected

Note: Claims labelled NEUTRAL are ungrounded — the model introduced information not in the retrieved context. The fix is to add more complete context chunks covering dosage and pharmacokinetics.

# Structured Fact-Check Tools vs LLM Self-Verification

## Structured Fact-Check Tools

- External APIs with authoritative data: Wolfram Alpha (math/science), Google Knowledge Graph, PubMed API, legal databases (Westlaw)
- Gold standard for numerical facts, scientific constants, and structured entity data
- Deterministic: same query always returns same verified answer
- Cannot verify subjective or nuanced claims
- Requires integration work per domain
- Best for: numerical claims, entity facts, scientific constants, legal citations, current prices/dates

## LLM Self-Verification (CoVe / Prompting)

- Model reviews its own claims independently — no external API needed
- General-purpose: works on any factual claim type
- Non-deterministic: model may hallucinate the same error in both generation and verification
- Anchoring risk: verification can be biased by the original draft if shown in context
- Cheaper to implement: no external API integration
- Best for: nuanced claims, opinion-grounded facts, cases where no structured API exists; always less reliable than external verification for numerical and entity facts

# Case Study — Medical RAG with Citation Enforcement

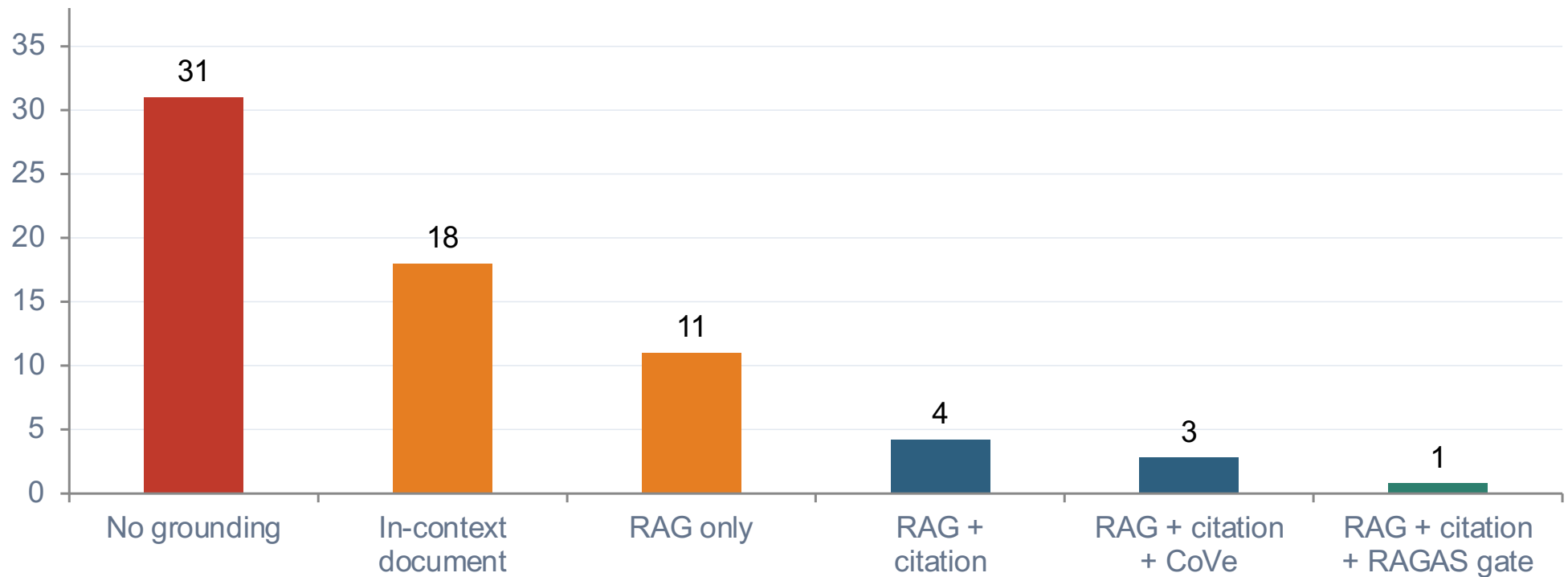
## Case: NHS Clinical Knowledge Summaries RAG System — Deployment Validation

- NHS England piloted a RAG-based clinical decision support tool grounded exclusively in CKS (Clinical Knowledge Summaries) — a curated evidence database of UK clinical guidelines
- Pre-RAG baseline: ungrounded GPT-4 on clinical queries showed 31% hallucination rate on drug dosage and contraindication questions
- Phase 1 — RAG only: reduced hallucination rate to 11% — but 11 in 100 clinical queries still contained an unsupported claim
- Phase 2 — RAG + citation enforcement: model required to cite specific CKS section for every claim. Hallucination rate fell to 4.2%
- Phase 3 — RAG + citation + RAGAS gate: faithfulness threshold 0.92 required before delivery; responses below threshold routed to pharmacist review. Patient-facing hallucination rate: 0.8%

## Key Lesson

Three-layer grounding (RAG + citation enforcement + RAGAS faithfulness gate) achieved 97.5% reduction from ungrounded baseline. The 0.8% residual rate corresponds to ~1 in 125 queries needing pharmacist intervention — fully covered by the human review pathway. Each grounding layer costs approximately \$0.003 per query additional overhead.

# Hallucination Rate Before vs After Grounding Strategies



*Source: NHS CKS pilot data (2024). Each grounding layer provides multiplicative reduction. The final RAGAS gate catches residual faithfulness failures that citation enforcement misses, bringing the patient-facing rate below 1%.*



# Key Takeaways

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- 1 Grounding constrains generation to verified external sources — it reduces hallucination by 5–7× in RAG configurations but does not eliminate faithfulness hallucinations (model misrepresenting its own sources)
- 2 The RAG pipeline has two independent failure modes: retrieval failure (wrong context retrieved) and faithfulness failure (model departs from retrieved context) — both must be measured and mitigated separately
- 3 Citation-enforced generation requires the model to attribute every factual claim to a numbered source — it reduces extrinsic hallucinations by 60–80% and makes unverifiable claims immediately visible
- 4 Chain-of-Verification (CoVe) improves factual accuracy 28–40% by having the model verify its own claims independently — valuable for high-stakes, low-volume queries where 3–4× query cost is justified
- 5 RAGAS faithfulness score is the standard quality gate: require  $\geq 0.90$  before production deployment; continuous daily monitoring on a 5% production sample; alert on any drop below 0.85